

Heat Transfer

Final Lab Report
Unit Operations Lab 2 
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Group 2

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Table of Contents

Executive Summary	1
Introduction.....	1
Theory	4
Results.....	7
Discussion.....	9
References.....	15
Appendix I: Data Tabulation/Graphs.....	16
Appendix II: Error Analysis.....	22
Appendix III: Sample Calculations.....	22
Appendix IV: Individual Team Contributions.....	26

Executive Summary

The purpose of this experiment was to determine the heat transfer coefficient (U) between the fluid (**water**) and inside the jacketed vessel. A set of two experiments were performed in order to study the effect of different factors including flow rate of the fluid (water), impeller speed, and presence of baffles on the heat transfer coefficient at steady state. The first experiment was done without the presence of baffles, and the impeller speed was varied at three different speeds (~ **650 RPM, 830 RPM, 1100 RPM**). It was found out that the energy done by the heat exchanger are larger than when the baffles were applied, also, same goes to the Pr, Re and Nu numbers are much larger than with the baffles in the tank. For instant, at the 650 RPM flow rate, the Nu number is 44213 and it drops by increasing the flowrate to 830 to be 1803. Another example is the Re number, where the range for the trial with baffles on was between (169076 to 284021). In addition, for the trial where the baffles were taken off, the Re number had a range between (36142 to 64286). The variation of the impeller speed changes all the heat transfer coefficients, for example with no baffles the heat transfer coefficient at 650 RPM was 2801.58 and with the increasing of the speed of the impeller rotation to 830 RPM, the heat transfer coefficient increased 3314.48. On the other hand, at the speed of 610 RPM with baffles on, the heat transfer coefficient was 984.62, and increased to 1169.86 when the speed of the impeller rotation increased. All these examples and numbers are going to be presented on the data/tabulation section of this report. The same process was repeated with the presence of baffles and without them several times with changing the flow rate and the speed of the impeller rotation to gather all the data needed to plot graphs and compare gathered data with theoretical ones

Introduction:

Heat transfer is defined as the study of the flow of heat, and it plays an important role in engineering field such as chemical, mechanical and civil engineering [1]. In fact, there are three types of heat transfer which are conduction, convection and radiation [2]. Heat transfer is considered conduction if transfers via solid material, and if it transfers via liquid or gas is defined as convection, or radiation if it transfers via electromagnetic waves. For the purpose of this experiment, the heat transfer is classified as convection since water is the used fluid.

Predicting the rate of heat transfer is significantly important to establish a variety of chemical engineering processes [3]. For instance, understanding the rate of heat transfer is an important factor in the usage of process steam, which is used mostly as heating utility in most plants. It is also a key to minimize the heat loss when transporting the steam through insulated steam pipes [3].

Heating and cooling of batch tank is another application of heat transfer in chemical process industries [4]. This application was used for the purpose of this experiment in which cooling water was used as cooling method, and steam was used as heating method. In this experiment, a stirred tank with recirculation stream, and internal cooling coil and adjustable baffle as shown in **figure 1** was used.

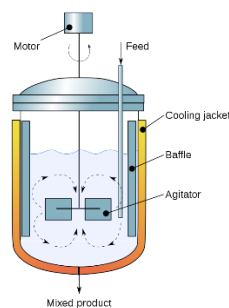


Figure 1: General Schematic of Stirred tank

In fact, stirred tank is considered an important equipment that is mainly used for mixing in many industries such as chemical, food and pharmaceutical. In addition to that, stirred tank can be used for the process of heating by using steam jacket, or cooling using cooling coil [5]. Thus, it provides the best system to analyze the process of heat transfer by measuring the heat transfer coefficient.

Heat transfer coefficient (U) is defined as a parameter used to determine convection of heat transfer between the fluid and the surface, which is the vessel's wall. The role of heat transfer coefficient is significant in order to properly design the internal heating or cooling devices [6]. In order to understand the heat transfer in a stirred tank, experiment as shown in figure 2 is needed to analyze the flow pattern since it is hard to analyze analytically [5].

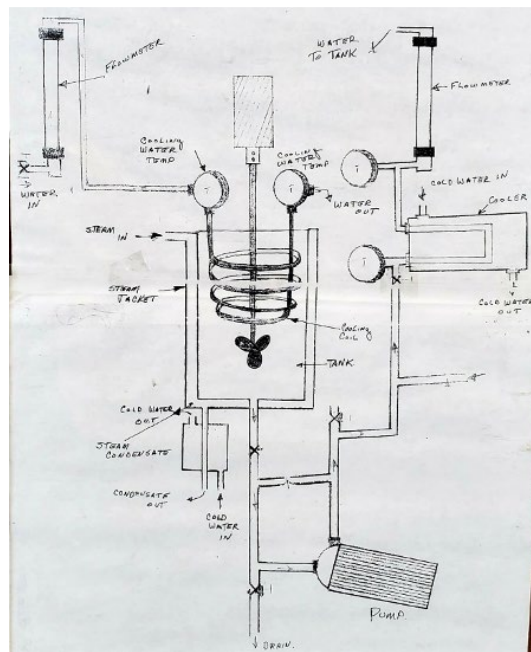


Figure 2: General schematic of the apparatus of the experiment.

Theory:

The inner wall heat transfer coefficient for unbaffled vessel containing Newtonian fluid, such as water used in this experiment, can be calculated using the following equation:

$$\frac{h_i D_T}{k} = 0.36 \left(\frac{D_i^2 n \rho}{\mu} \right)^{\frac{2}{3}} \left(\frac{C_p \mu}{k} \right)^{\frac{1}{3}} \quad (\text{equ 1})$$

Where:

- h_i = convective heat transfer coefficient at the inner surface of the tank (W/m² K)
- D_T = tank diameter (m)
- μ = viscosity of the fluid (kg/m s)
- C_p = specific heat of the fluid (J/ kg K)
- k = thermal conductivity of the fluid (W/ m K)
- ρ = density of the fluid (kg/ m³)
- D_i = impeller diameter (m)
- n = speed of rotation of the impeller, (revolutions per minute (RPM))

The dimensionless groups for the correlation of the difference in temperature between the inside vessel wall and the bulk mean temperature of the well stirred fluid are:

NUSSELT NUMBER (TANK)

$$\frac{h D_T}{k} = Nu_T$$

REYNOLDS NUMBER (IMPELLER)

$$\frac{D_i^2 n \rho}{\mu} = Re_{imp}$$

PRANDTL NUMBER (FLUID)

$$\frac{C_p \mu}{k} = Pr$$

Inserting the dimensionless numbers into Eq.1, it can be written as following

$$Nu_T = 0.36 (Re_{imp})^{\frac{2}{3}} (Pr)^{1/3} \quad (\text{equ2})$$

Film condensation from the steam jacket used for the purpose of heating occurs on the outside of the vessel wall. The following equation is used in order to determine the **outside heat transfer coefficient**,

$$\frac{h_o L_T}{k} = 0.925 \left(\frac{L^3 \rho^2 g}{\mu \Gamma} \right)^{1/3} \quad (\text{equ3})$$

Where

- h_o : the hydrothermal carbonization (HTC) through the condensing film (W/m²*K)
- L_T : vertical length of the tank
- ρ : density of the condensed steam
- g : the gravity acceleration

- μ : Viscosity of the condensed steam
- Γ : mass rate of steam condensation per wetted perimeter, and it can be calculated using

$$\Gamma = \frac{\dot{m}_{st}}{\pi D_{T,o}}$$

Where

- $\dot{m}_{st}$: the mass rate of steam condensation (kg/s)
- $D_{T,o}$: the outer diameter of the tank (m)

***** Note:** Equ 3 is only valid of $Re_{imp} < 2100$, and it can be calculated using the following

$$Re_{imp} = \frac{4\Gamma}{\mu}$$

The HTC through the condensing film (h_o) for **saturated steam at 1 atm** can be evaluated using the following equation

$$h_o = 2960 \left(\frac{D_{T,o}}{\dot{m}_{st}} \right)^{1/2} \quad (\text{equ 4})$$

In order to calculate the heat transfer flux (Q) at steady state for the interface of liquid-solid, the following equation is used

$$Q = h_i A_i (t_4 - t_5) = h_o A_o (t_1 - t_2) \quad (\text{equ 5})$$

Where

- h_i : the film heat transfer at the inner surface
- h_o : the film heat transfer at the outer surface
- A_i : area of the inner surface (m^2)
- A_o : area of the outer surface (m^2)
- t_1 : temperature of the hot fluid (K)
- t_2 : temperature of the inner wall (K)
- t_4 : temperature of the outer wall (K)
- t_5 : temperature of the cool fluid (K)

***** Note:** For steady state, the total heat flux is constant, and there is no accumulation of heat. The heat flux (Q) for each individual resistance have to be the same, and the sum of all the individual must be equal to the total heat flux across the interface.

Fourier's law is used in order to determine the heat transported through two solid surfaces, and it is expressed as following

$$Q = k A \frac{dt}{dx} = \left[A_{scale} k_{scale} \frac{(t_2 - t_3)}{x_{scale}} = A_{wall} k_{wall} \frac{(t_3 - t_4)}{x_{wal}} \right] \quad (\text{equ 6})$$

Where

- A_{scale} : the thickness of the scale
- A_{wall} : the thickness of the wall
- k_{scale} : the thermal conductivity of the scale
- k_{wall} : the thermal conductivity of the wall

In terms of local interface temperatures, heat transfer across each resistance can be expressed as following if the inner area of the wall is considered

$$Q = U_i A_i (t_1 - t_5) = [U_i A_i [(t_1 - t_2) + (t_2 - t_3) + (t_3 - t_4) + (t_4 - t_5)]] \quad (\text{equ7})$$

Where

- U_i : the total heat transfer coefficient ($\text{W}/\text{m}^2 \text{K}$)

Local interface temperature differences are a function of total heat flux and, local heat transfer coefficient, or conductivity and thickness of the solid resistance

For instance:

- $t_1 - t_2 = \frac{Q}{h_o A_o}$
- $t_2 - t_3 = \frac{Q_{x,\text{scale}}}{A_{\text{scale}} k_{\text{scale}}}$

by substituting and simplifying, the overall transfer coefficient (U_i) of the overall resistance for resistors in series can be expressed as following:

$$U_i = \frac{1}{\left(\frac{1}{h_i}\right) + \left(\frac{x_{\text{scale}}}{k_{\text{scale}}} \frac{A_i}{A_{\text{scale}}}\right) + \left(\frac{x_{\text{wall}}}{k_{\text{wall}}} \frac{A_i}{A_{\text{wall}}}\right) + \left(\frac{1}{h_o} \frac{A_i}{A_o}\right)} \quad (\text{equ 8})$$

Where

U_i : the overall heat transfer coefficient ($\text{W}/\text{m}^2 \text{K}$)

For the purpose of this experiment, the overall transfer coefficient (U_i) equation can be simplified to the following equation:

$$U_i = \frac{1}{\left(\frac{1}{h_i}\right) + \left(\frac{1}{h_o}\right)} \quad (\text{equ 9})$$

by assuming the following based on **figure 3**

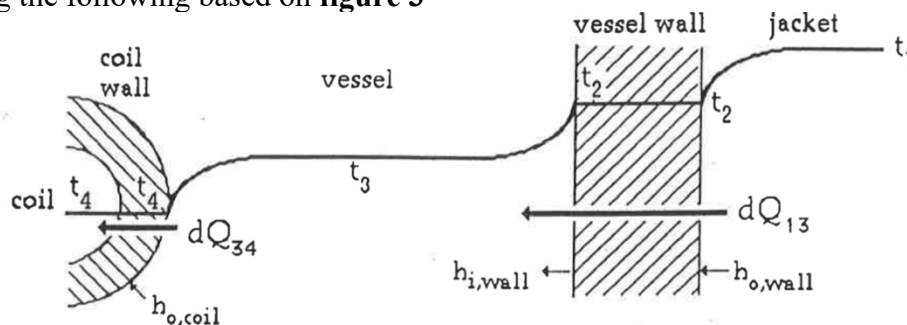


Figure 3: simplified scenario for calculation of U_i

Assumptions:

- There is no scale of either side of solid wall
- Temperature gradient across solid wall is small; thus, it's negligible
- The ratio $\frac{A_i}{A_o} \approx 1$ since the thickness of the wall is small compared to diameter of tank
- Temperature is uniform since the liquid in the vessel assumed well-mixed

Therefore,

$$U_i = \frac{1}{\left(\frac{1}{h_i}\right) + \left(\frac{1}{h_o}\right)} \quad (\text{equ 9})$$

And

$$Q_{13} = U_{i,\text{wall}}(t_1 - t_3)A \quad (\text{equ 10})$$

Where:

- t_1 : temperature of steam inside the jacket (K)
- t_3 : the average bulk temperature of the fluid inside the tank (K)
- A : the heat transfer surface area (m^2)
- $U_{i,\text{wall}}$: the overall heat transfer coefficient of the vessel wall ($\text{W}/\text{m}^2 \text{ K}$)
- Q_{13} : heat transferred (J/s)

Furthermore, the heat transferred from the steam to the tank (Q_{13}) can be established via the energy balance on a constant volume among steam jacket and steam condenser as the following equation show:

$$Q_{13} = m_{\text{st}}(h_{\text{st,in}} - h_{\text{st,out}}) - Q_{\text{sc}} - Q_{\text{surr}}$$

Where:

- $h_{\text{st,in}}$: the enthalpy of steam in
- $h_{\text{st,out}}$: the enthalpy of steam out
- Q_{sc} : the heat removed in the steam condenser
- Q_{surr} : the heat loss of the insulated vessel to the surroundings

A heat Balance around the jacketed vessel is **another method** to determine the transferred heat as shown below

$$\begin{aligned}
 (\text{Heat In}) &= (\text{Heat Out}) \\
 Q_{13} &= Q_{\text{hx}} + Q_{\text{c}} + Q_{\text{suur}}
 \end{aligned}$$

Where

- Q_{13} : heat transferred (J/s)
- Q_{hx} : the amount of removed heat in the external heat exchanger
- Q_{c} : the amount of removed heat by cooling coils
- Q_{surr} : the heat loss from the recirculation pipes

Where the equation become the following at **steady state**,

$$Q_{13} = Q_{\text{hx}} + Q_{\text{c}}$$

The amount of heat removed in the external heat exchanger (Q_{hx}) can be calculated using the flowrate and the temperature drop of the fluid recirculated through heat exchanger as the following equation explains,

$$Q_{\text{hx}} = m_{\text{hx}} C_p(t_3 - t_r)$$

Where

- m_{hx} : the amount of fluid circulated through the heat exchange (kg/s)
- C_p : the heat capacity of the fluid ($\text{J}/\text{kg K}$)
- t_3 : temperature of the fluid entering the heat exchanger (K)

- t_r : temperature of the fluid leaving the heat exchanger and returning to the tank (K)

Similar method can be used to determine the amount of heat removed by the cooling coils (Q_c) using both the flowrate of the water in the coils and temperature rise of water passing the coils as shown in the following equation:

$$Q_c = m_c C_p (t_{out} - t_{in})$$

Where

- m_c : the flowrate of water through the cooling coils
- C_p : the heat capacity of water (J/kg K)
- $(t_{out} - t_{in})$: the temperature difference of the cooling water inside the coils. (K)

Results

The main objective of this experiment was to observe the relation between the heat transfer of the heated jacket and cooling coil with a propeller for agitation; specifically using 650, 800, 1100 RPM for the impeller while keeping the temperature constant to compare the dimensionless variables. Furthermore, two different set ups were used one with baffles and the other without, the two setups are used to determine if a different property parameter, increased mixing, would differ the heat transfer coefficient. Examining what occurs will help understand what occurs in reactors, heat exchangers, or containers. The experiments were taken under steady state conditions.

Observing the results, it is found that the relation of the Reynolds number and the ratios of the Nusselt and Prandtl number are related in a linear trend, meaning that as the flow was increased the dimensionless variables were increasing as well making it relevant. As seen in the results below in **figure 4**.

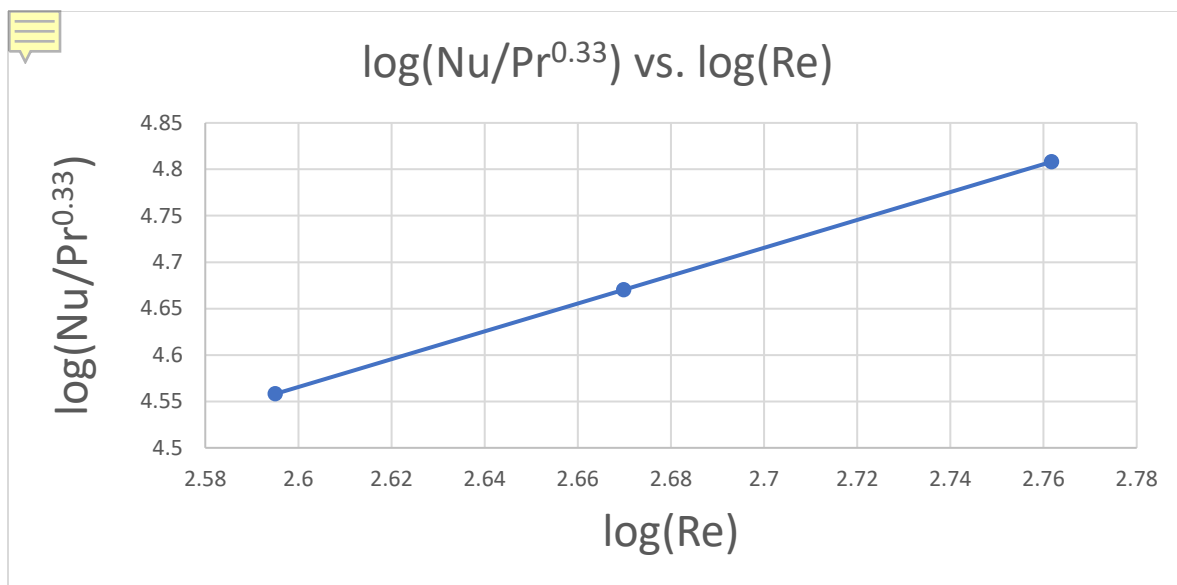


Figure 4: Nusselt number over Prandtl vs the log of Reynolds number (Baffles on)

Furthermore, observing the trends of the impeller speed with the baffles and unbaffled set ups, it is seen on **figure 5** that the unbaffled inner heat transfer coefficients are higher in the unbaffled setup compared to the low baffled heat transfer coefficient, the lower baffled meaning that having less turbulence or mixing is better for a higher inner heat transfer coefficient. The data collected could be used in industry to understand what situation best fits for inner heat transfer when choosing a vessel like a CSTR. The data collected for inner heat transfer coefficients vs impeller speed can be seen below in **figure 5**.

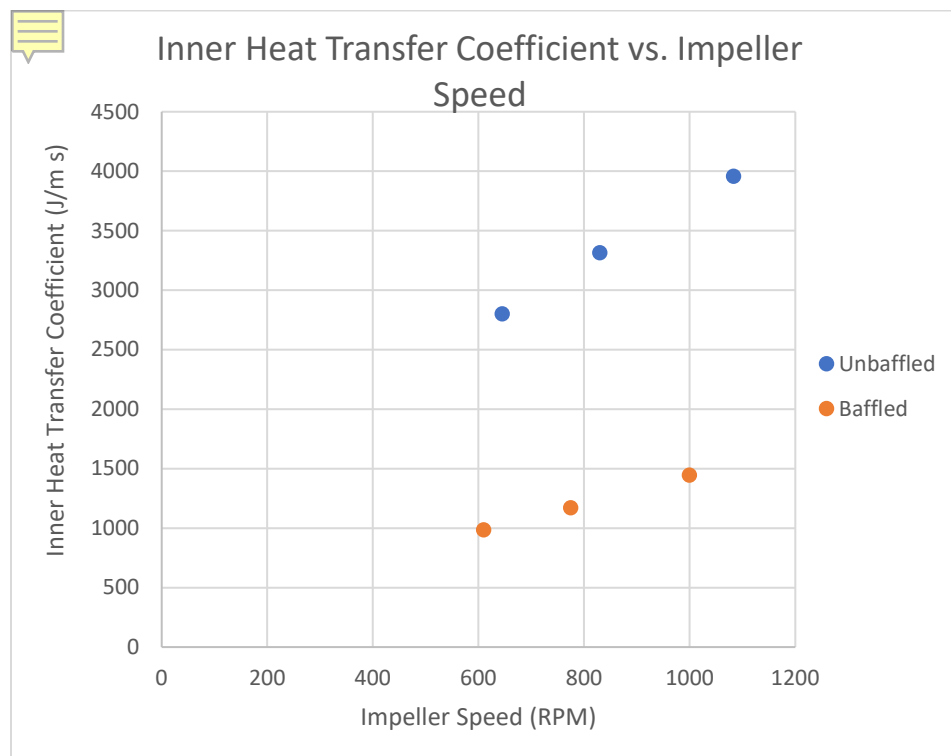


Figure 5: Heat transfer coefficient vs Inner heat transfer coefficient

Henceforth, observing the data collected for overall heat transfer coefficient for both the unbaffled and baffled vessel, it is found that as the impeller speed increased the heat transfer coefficient drops close to 10 J/m*s then the trend seems to be increasing steadily as the speed

increased. While, for the baffled setup the trend is opposite although it is increasing with a similar slope. The overall heat transfer coefficient trends can be seen on figure 6 below.

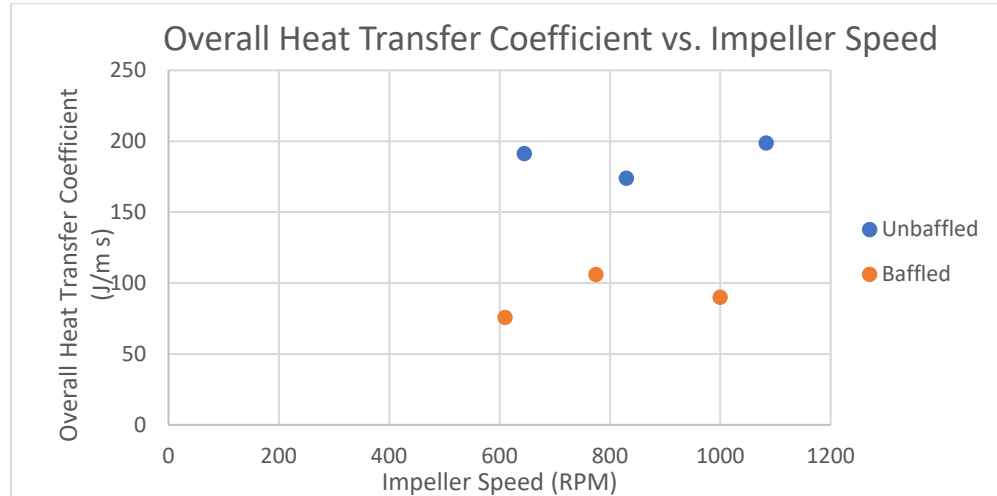


Figure 6: Overall heat transfer coefficient vs Impeller speed

In addition, observing the results for no baffles, it is found that the relation of the Reynolds number and the ratios of the Nestle and Prindles number are related in a linear trend, meaning that as the flow was increased the dimensionless variables were increasing as well making it relevant. As seen in the results below in **figure 7**.

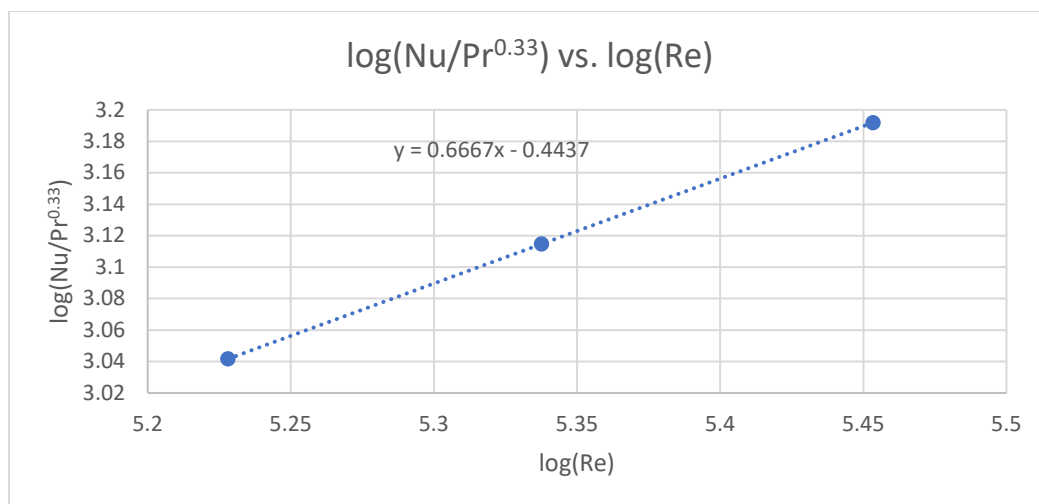


Figure 7: log (Nu/Pr^{0.33}) vs. log(Re) for no baffles

Lastly, observing the results in **figure 8** it is found that there is a higher heat transfer at lower cooling water flowrates. It also demonstrates that at higher cooling water flowrates whether the tank is baffled or not, does not affect the heat transfer since they are relatively similar, but when the flow rate is lower the unbaffled the heat transfer coefficient is higher by 55 J/(m*s). Thus, having the unbaffled low flow rate of cooling water results in that the heat transfer in the tank to be larger, an example that this result that can be useful in the industry could be when using a batch reactor that is exothermic and needing to cool the tank.

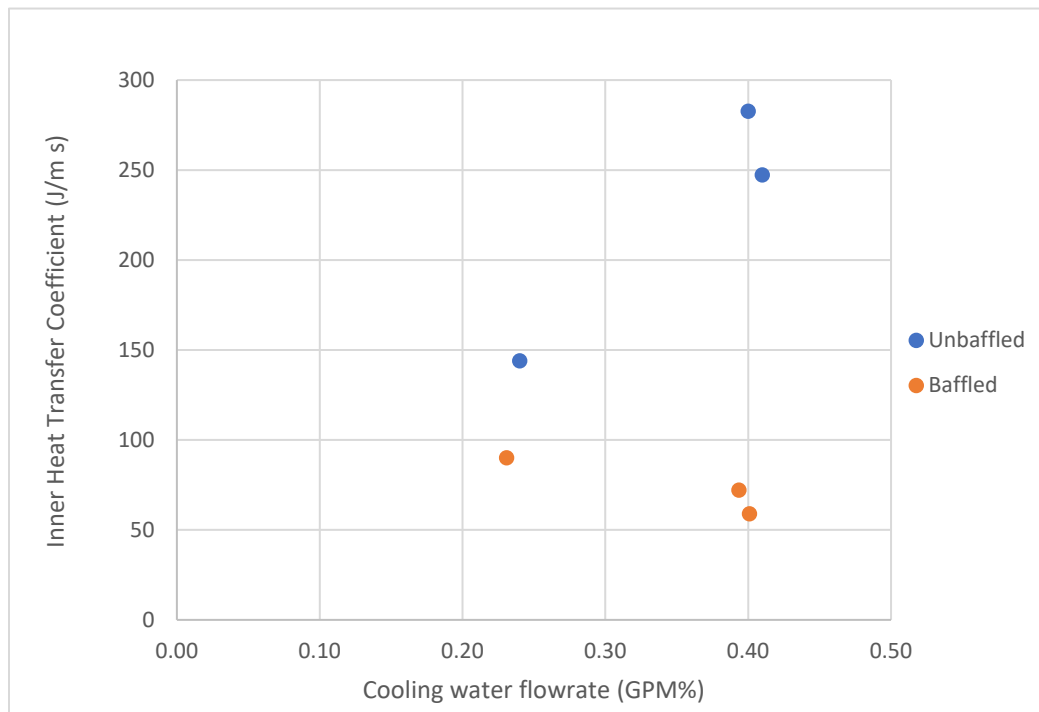


Figure 8: Inner Heat Transfer Coefficient vs Cooling Water Flowrate at (1000RPM)

Discussion

In this experiment was to determine the heat transfer coefficient among the fluid by the wall and the fluid in the tank. The stir in the CSTR tank varied between 650 to 1100 rpm, follow-on temperature reading of the fluid on the system. The heat transfer coefficient is dependent on the flow parameters properties and the physical properties of the what the tank is containing. The correlation development of the experimental data can lead to determine the heat transfer quantities under some conditions.

The stir in the CSTR ensured the tank to be evenly distributed. It projects the heat transfer is proportional to the difference in the temperature between the fluid by the wall and elsewhere. The greater the temperature difference the greater is the heat transfer and it goes for opposite condition when the lower the temperature difference the lower is the heat transfer. In case such as, a batch tank when there is no mixer the temperature between the surface and the fluid will equilibrate eventually.

The baffles are to perform a well mix solution in the CSTR. Furthermore, it increases the heat transfer area in the CSTR. It would be assumed that the inner heat transfer coefficient to be larger with this attachment (baffles).

If one to compare between the overall heat transfer coefficient with the speed impeller, the overall heat transfer is higher when the baffles was preforming. On the other hand, if the baffles were not working the heat transfer are lower in this case. As shown in the results the temperature variable is dependent upon of the heat transfer coefficient. The increase of temperature in the tank the inner heat transfer proportionally.

Some experimental difficulties were to maintain the flow at steady state. The flow of the cold water constantly decreasing with time. One of the members kept manually changing the

flow to reach it steady state. An assumption was made that the bulk properties of the water in the CSTR was constant.

Setting both heat exchangers at steady state flow to allow the temperature inlet flow and outlet flow was used to analyze the heat flow system. The difference of the heat quantities that entering the system and out of the system was used to determine for the experimental data for this procedure.



References:

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Appendix I: Data Tabulation/Graphs

Table 1 Experiment A Raw Data

Experiment A: No baffles								
(heat change as a function of agitation)				units		(heat change as a function of flowrate)		
RPM	640-650	825-835	1073-1094			RPM	1073-1094	1073-1094
T-1	278.15	278.15	278.15	C		T-1	278.15	277.59
T-2	282.59	278.98	279.26	C		T-2	280.37	282.04
T-3	277.59	277.59	277.59	C		T-3	277.59	277.04
T-4	322.04	323.15	327.04	C		T-4	316.48	316.48
T-5	338.71	338.15	338.71	C		T-5	338.71	333.15
T-6	322.15	335.15	325.15	C		T-6	395.15	402.35
T-7	337.15	319.15	340.15	C		T-7	417.65	418.55
F1	24.8	23.9	24	GPM		F1	40	41
F2	21	19	16.5	GPM		F2	40	55
Time (min)	10.67	10.96	10.86			Time (min)	10.53	10.65
Volume of condensate (mL)	35.7	32.5	66			Volume of condensate (mL)	48.8	34

Table 2 Experiment B Raw Data

Experiment B: With baffles								
(heat change as a function of agitation)					(heat change as a function of flowrate)			
RPM	610-621	790-800	1073-1094		RPM	1070-191	1080-193	
T-1	277.59	278.15	278.71		T-1	277.59	277.59	
T-2	320.37	279.26	278.15		T-2	311.48	307.59	
T-3	278.15	277.59	277.59		T-3	277.59	278.15	
T-4	320.71	320.37	318.15		T-4	278.15	279.82	
T-5	333.71	332.59	327.59		T-5	333.15	327.59	
T-6	298.15	299.15	309.15		T-6	318.05	321.15	
T-7	314.65	329.15	328.05		T-7	332.05	332.15	
F1	0.23	0.23	0.231		F1	40.1	39.5	
F2	7	7	7		F2	41	60	
Time (min)	10.65	10.62	10.53		Time (min)	10.53	10.41	
Volume of condensate (mL)	70.5	69.5	109.5		Volume of condensate (mL)	52.9	21.2	

Table 3 Physical Properties used for calculations

Heat Capacity (Cp) HX	Viscosity (mu) HX [kg/m*s]	Density (rho) HX	Thermal Conductivity (k) HX	Heat Capacity (Cp) COIL	Viscosity (mu) COIL	Density (rho) COIL	Thermal Conductivity (k) COIL
4184	0.0004	983	0.6305	4184	0.0018	999.842	0.561

Table 4 Summary of Values for Unbaffled Tank (Experiment A)

	T-3	T-4	F1	T-7	T-6	F2	Stem Temp	T-2	
Trial	Impeller Speed [rpm]	Cooling Water Temp In [K]	Cooling Water Temp Out [K]	Cooling Water Flowrate [GPM%]	Vessel Fluid Temp into Tank (from Exchanger)	Vessel Fluid Temp Out of Tank (in Vessel)	Vessel Fluid Flowrate [GPM%]	Steam Temp (t1)	Condensate Temp
Low Speed	645.00	277.59	322.04	0.25	322.15	337.15	0.21	373.15	255.37
Middle Speed	830.00	277.59	323.15	0.24	319.15	335.15	0.19	373.15	278.98
High Speed	1083.50	277.59	327.04	0.24	325.15	340.15	0.17	373.15	279.26
	1083.00	277.59	316.48	0.40	417.65	395.15	0.40	373.15	280.37
	1083.00	277.04	316.48	0.41	418.55	402.35	0.55	373.15	282.04
Total GPM				0.94			1.1		

Table 5 Summary of Values for Baffled Tank (Experiment B)

	T-3	T-4	F1	T-7	T-6	F2	Stem Temp	T-2	
<i>Trial</i>	Impeller Speed [rpm]	Cooling Water Temp In [K]	Cooling Water Temp Out [K]	Cooling Water Flowrate [GPM%]	Vessel Fluid Temp Into Tank (from Exchanger)	Vessel Fluid Temp Out of Tank (in Vessel)	Vessel Fluid Flowrate [GPM%]	Steam Temp (t1)	Condensate Temp
<i>Low Speed</i>	610	278.15	320.71	0.23	314.65	298.15	0.07	386.15	320.37
<i>Middle Speed</i>	790	277.59	320.37	0.23	329.15	299.15	0.07	386.15	279.26
<i>High Speed</i>	1085	277.59	318.15	0.23	328.05	309.15	0.07	386.15	278.15
	1191	277.59	278.15	0.40	332.05	318.05	0.41	386.15	311.48
	1093	278.15	279.82	0.39	332.15	321.15	0.60	386.15	307.59
<i>Total GPM</i>				0.94			1.1		

Table 6 Calculated Values for Baffled Tank (Experiment A)

Impeller Speed [rpm]	Q _{hx} (Heat Exchanger/ Recirculation) [kW]	Q _{coils} (Cooling Water) [kW]	Q ₁₃ [kW]	Tank Nusselt Number (Nu _T)	Impeller Reynolds Number (Re _I)	Fluid Prandtl Number (Pr)	h(inner)	U(wall)	h(outside)	Rotations per sec
645	0.90	2.73	3.63	1524.10	169076.00	2.65	2801.58	130.16	136.50	10.75
830	0.87	2.70	3.57	1803.12	217570.67	2.65	3314.48	120.73	125.30	13.83
1083.5	0.71	2.94	3.65	2153.73	284021.47	2.65	3958.98	138.93	143.98	18.06
1083	2.57	3.86	6.43	2153.07	283890.40	2.65	3957.76	263.88	282.74	18.05
1083	2.54	4.01	6.56	2153.07	283890.40	2.65	3957.76	-263.78	247.29	18.05

Table 7 Calculated Values for Baffled Tank (Experiment B)

Impeller Speed [rpm]	Q _{hx} (Heat Exchanger/ Recirculation) [kW]	Q _{coils} (Cooling Water) [kW]	Q ₁₃ [kW]	Tank Nusselt Number (Nu _T)	Impeller Reynolds Number (Re _I)	Fluid Prandtl Number (Pr)	h(inner)	U(wall)	h(outside)	Rotations per sec
610	0.33	2.43	2.75	535.65	36142.44	2.52	984.62	70.39	75.81	10.17
										13.17
775	0.59	2.44	3.03	636.42	46807.42	2.52	1169.86	97.27	106.09	
1000	0.37	2.32	2.70	786.34	64286.14	2.52	1445.45	84.84	90.13	18.08

1191	1.62	0.06	1.68	836.76	70566.63	2.52	1538.12	56.68	58.85	19.85
1093	1.87	0.16	2.03	790.20	64760.14	2.52	1452.54	68.65	72.06	18.22

Appendix II: Error Analysis

Throughout the experiment, there were some rooms of errors to occur. For instance, unexpected leak in the pump occurred, which considered a mechanical issue. Although such an issue may affect the flowrate reading of the water, yet the effect was too small to be notified. Therefore, the data obtained still considered precise and reliable. Another mechanical issue occurred during the experiment; the valve used to adjust the flow rate of cooling water as considered loose as it slightly moves by itself. Therefore, the reading of water flow rate was scientifically impacted by the issue as the data was considered not accurate. Therefore, repeating the experiment was required as one team member had to keep holding the valve in order to prevent the issue to occur.

Since multiple students were reading the rotameter to obtain the flowrate of cooling water, personal uncertainty may have occurred. The reason for that is personal judgment as reading the rotameter vary from someone's eyes to another's which decrease the accuracy of the data obtained. Besides reading the rotameter, personal judgment may have also occurred while reading the dial thermometers, tachometer due to the same reason. Furthermore, small difference of ($\sim 0.1 - 0.5$) may have affected reading the volume of the collected condensate inside the graduated cylinder due to personal uncertainty. Another example of personal judgment is measuring the time of the collecting the condensate. The time was supposed to be ~ 10 s. However, due to personal judgment, a small change of ($0.1 - 0.6$).). Personal uncertainty or judgment is one type of a systematic errors [8]. Although there were some small uncertainties, the accuracy of the obtained data was not affected.

Appendix III: Sample Calculations

- Reynold's Number of the impeller (Re_{imp}):**

$$Re_{imp} = \frac{D^2 n \rho}{\mu} = \frac{(0.08)^2 (10.8)(980.45)}{(0.000434)}$$

$$Re_{imp} = 156000$$

- D: diameter of the impeller (m)
- n : revolutions per minute of the impeller (rev/s)
- ρ : density of liquid (kg/m³)
- μ : viscosity of the fluid (kg/m*s)

- Prandtl Number (Pr):**

$$Pr = \frac{C_p \mu}{k} = \frac{(4184)(0.000434)}{(0.6305)}$$

$$Pr = 2.88$$

where

- C_p : heat capacity of water (J/kg*K)
- μ : Viscosity of the fluid (kg/m*s)
- k: thermal conductivity of the fluid (w/m*K)

- Nusselt Number (Nu):**

$$Nu_T = 0.36(Re_{imp})^{2/3}(Pr)^{1/3} = .36(156000)^{2/3}(2.88)^{1/3}$$

$$Nu_T = 1485$$

- Convective heat transfer coefficient in the tank (h_i) [W/m² * K]**

$$h_i = \frac{Nu_T * k}{D_T} = \frac{1485 * 0.6305}{0.343}$$

$$h_i = 2700 \frac{J}{s * K}$$

where

- k: Thermal conductivity of the fluid (J/s*m*K)
- D_T : Diameter of the tank (m)

- Calculating the mass flow of water through the heat exchanger (m_{h_v}) [kg/min]:**

$$m_{hx} = V_h \cdot \rho = \frac{(0.20 \cdot 1.1) \text{ gal}}{\text{min}} \cdot \frac{1 \text{ m}^3}{264.172 \text{ gal}} \cdot \left(\frac{983.3 \text{ kg}}{1 \text{ m}^3} \right)$$

$$m_{hx} = 0.82 \frac{\text{kg}}{\text{min}}$$

where

- V_h : the volumetric flow rates of hot water (gal/ min)
- ρ : density of the hot water (kg/ min)

- **Calculating the mass flow of fluid cold water through the vessel (m_c) [kg/min]**

$$m_c = V_c \cdot \rho = \frac{(0.24 \cdot 0.94) \text{ gal}}{\text{min}} \cdot \frac{1 \text{ m}^3}{264.172 \text{ gal}} \cdot \left(\frac{998.8 \text{ kg}}{1 \text{ m}^3} \right)$$

$$m_c = 0.85 \frac{\text{kg}}{\text{min}}$$

where

- V_c : the volumetric flow rates of cold water (gal/ min)
- ρ : density of the hot water (kg/ min)
- **The amount of heat removed in the external heat exchanger (Q_{hx})**

$$Q_{hx} = m_{hx} C_p (t_3 - t_r)$$

$$Q_{hx} = \left(0.82 \frac{\text{kg}}{\text{min}} \right) \left(\frac{\text{min}}{60 \text{ s}} \right) \left(4184 \frac{\text{J}}{\text{kg K}} \right) (337.15 \text{ K} - 322.15 \text{ K})$$

$$Q_{hx} = 0.90 \frac{\text{J}}{\text{s}}$$

Where

- m_{hx} : the amount of fluid circulated through the heat exchange (kg/s)
- C_p : the heat capacity of the fluid (J/ kg K)
- t_3 : temperature of the fluid entering the heat exchanger (K)
- t_r : temperature of the fluid leaving the heat exchanger and returning to the tank (K)

- **Calculating the amount of heat removed by the cooling coils (Q_c)**

$$Q_c = m_c C_p (t_{out} - t_{in})$$

$$Q_c = \left(0.85 \frac{\text{kg}}{\text{min}} \right) \left(\frac{\text{min}}{60 \text{ s}} \right) \left(4184 \frac{\text{J}}{\text{kg K}} \right) (322 \text{ K} - 255 \text{ K})$$

$$Q_c = 4.10 \frac{\text{J}}{\text{s}}$$

Where

- m_c : the flowrate of water through the cooling coils
- C_p : the heat capacity of water (J/kg K)
- $(t_{out} - t_{in})$: the temperature rise of the cooling water inside the coils/. (K)

- **the heat transferred from the steam to the tank at steady state (Q_{13}):**

$$Q_{13} = (Q_{hx} + Q_c) = (0.90 \text{ J/s}) + (4.10 \text{ J/s})$$

$$Q_{13} = 5.00 \frac{\text{J}}{\text{s}}$$

Where:

- Q_{hx} : The amount of heat removed in the external heat exchanger (J/s)
- Q_c : Calculating the amount of heat removed by the cooling coils (J/s)

- **The overall heat transfer coefficient ($U_{i,wall}$):**

$$U_{i,wall} = \frac{Q_{13}}{(t_1 - t_3)A} = \frac{(5.00 \frac{\text{J}}{\text{s}})}{(373.15 \text{ K} - 325.15 \text{ K})(0.547 \text{ m}^2)}$$

$$U_{i,wall} = 0.40 \frac{\text{W}}{\text{m}^2\text{K}}$$

Where:

- t_1 : temperature of steam inside the jacket (K)
- t_3 : the average bulk temperature of the fluid inside the tank (K)
- A : the heat transfer surface area (m^2)
- Q_{13} : heat transferred (J/s)

Appendix IV: Individual Team Contributions

NAME: Ahmad Abusubaih

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	3	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	3	
Materials & Supplies (prelab)	1	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	3	
Executive Summary (final lab)	0	
Introduction (Final lab)	0	
Theory (Final Lab)	1	
Results (final lab)	1	
Discussion (final lab)	3	
Conclusion (final lab)	0	
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	2	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	29	

NAME: **Fatimah Al Ibrahim**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)	1	
Introduction (Final lab)	1	
Theory (Final Lab)	1	
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	1	
Power Point Presentation	1	
Total	22	

NAME: Ahmed Alghamdi

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	1	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	1	
Materials & Supplies (prelab)	1	
Procedure (prelab)	1	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)	1	
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	1	
Discussion (final lab)	1	
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	19	

NAME: **Velia Palma**

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	8	
Pre-Lab Editing	3	
Experimental Objective (Prelab)	3	
Apparatus (prelab)		
Materials & Supplies (prelab)	1	
Procedure (prelab)	2	
Project Safety Evaluation (prelab)		
Final Lab Editing	1	
Executive Summary (final lab)	1	
Introduction (Final lab)		
Theory (Final Lab)	2	
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	3	
Error Analysis (final lab)		
Sample Calculations (final Lab)	3	
Power Point Presentation		
Total	27	

NAME: Christian Sanchez

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	5	
Pre-Lab Editing	2	
Experimental Objective (Prelab)	1	
Apparatus (prelab)	2	
Materials & Supplies (prelab)	2	
Procedure (prelab)	2	
Project Safety Evaluation (prelab)	2	
Final Lab Editing	2	
Executive Summary (final lab)	2	
Introduction (Final lab)	1	
Theory (Final Lab)	1	
Results (final lab)	3	
Discussion (final lab)	1	
Conclusion (final lab)	1	
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	1	
Error Analysis (final lab)	1	
Sample Calculations (final Lab)	1	
Power Point Presentation	0	
Total	29	